

Why Lasso Creates Sparsity

Contents

1. Overview	2
2. Reminder: the Ridge slope	2
3. The Lasso slope	2
3.1 The loss	3
3.2 Why cases are needed	3
3.3 Case $m > 0$	3
3.4 Case $m < 0$	3
3.5 Case $m = 0$	4
4. Watching the slope reach 0	4
4.1 Why Lasso reaches 0	5
4.2 Why Lasso stops at 0	5
4.3 Why Ridge never reaches 0	5
5. The dead zone	6
5.1 Ten features at once	7
5.2 The picture: a diamond and a circle	7
6. Checking with scikit-learn	8
7. Summary	10
8. Sources	10
9. Key terms	10

Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Regularisation ([Note 63](#)).
- **Leads to:** Elastic Net ([Note 69](#)).
- **Compare with:** Ridge regression ([Note 66](#)); L1 and L2 regularisation in neural networks ([Note 1026](#)).

1. Overview

Key point: In the Lasso formula for one feature, λ is subtracted from the top of the fraction, so a large enough λ makes the slope exactly 0. In Ridge, λ is added to the bottom, which can only make the slope small.

The Lasso Note showed that Lasso sets coefficients to exactly 0, while Ridge only shrinks them. A model in which many coefficients are exactly 0 is called **sparse**, so the effect is called **sparsity**.

“Why does Lasso create sparsity, and Ridge does not?” is one of the most common interview questions on regularisation. This Note answers it with the formula for the slope when there is one **feature** (an input variable, one column of the data table). The same idea carries over to many features.

2. Reminder: the Ridge slope

Key point: Ridge: $m = S / (D + \lambda)$. λ sits in the denominator.

To keep the formulas short, write

$$S = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad D = \sum_{i=1}^n (x_i - \bar{x})^2$$

D is always positive. S is positive when y tends to rise with x , and negative when it tends to fall.

- **Linear regression** (OLS Note): $m = S/D$ and $b = \bar{y} - m\bar{x}$.
- **Ridge** (Ridge maths Note): $m = S/(D + \lambda)$, with the same b .

3. The Lasso slope

Key point: Because of the absolute value, the derivative is different for $m > 0$ and $m < 0$, so the formula comes in cases.

3.1 The loss

Key point: The loss is the squared error plus $2\lambda|m|$; the 2 only keeps the result tidy.

With one feature, the Lasso loss is

$$L = \sum_{i=1}^n (y_i - mx_i - b)^2 + 2\lambda|m|$$

The factor 2 does not change the idea: it only rescales λ , and it makes the final formula simpler.

The penalty does not contain b , so differentiating with respect to b gives the OLS result again: $b = \bar{y} - m\bar{x}$. Substituting it, each error becomes $(y_i - \bar{y}) - m(x_i - \bar{x})$.

3.2 Why cases are needed

Key point: $|m|$ has no derivative at $m = 0$, but on each side of 0 it is a simple line.

The absolute value $|m|$ has a sharp corner at $m = 0$ (the Lasso Note's loss curves), so it cannot be differentiated there. On either side, though, it is simple:

- if $m > 0$, then $|m| = m$;
- if $m < 0$, then $|m| = -m$.

So we solve the two cases separately.

3.3 Case $m > 0$

Key point: For a positive slope, $m = (S - \lambda) / D$.

With $|m| = m$, setting the derivative to zero:

$$\frac{\partial L}{\partial m} = -2 \sum_{i=1}^n (x_i - \bar{x}) [(y_i - \bar{y}) - m(x_i - \bar{x})] + 2\lambda = -2S + 2mD + 2\lambda = 0$$

$$m = \frac{S - \lambda}{D}$$

3.4 Case $m < 0$

Key point: For a negative slope, $m = (S + \lambda) / D$.

With $|m| = -m$, the penalty term's derivative is -2λ instead of $+2\lambda$, so

$$m = \frac{S + \lambda}{D}$$

3.5 Case $m = 0$

Key point: When neither formula gives an answer with the right sign, the slope is 0.

The result of case 3.3 is only valid if it really is positive, which needs $S > \lambda$. The result of case 3.4 is only valid if it really is negative, which needs $S < -\lambda$. When S is between $-\lambda$ and λ , neither case applies, and the lowest point of the loss is the corner itself: $m = 0$.

Condition	Lasso slope
$S > \lambda$	$m = (S - \lambda)/D$
$-\lambda \leq S \leq \lambda$	$m = 0$
$S < -\lambda$	$m = (S + \lambda)/D$

4. Watching the slope reach 0

Key point: Subtracting λ from S brings the top of the fraction down to exactly 0. After that, the slope stays at 0.

Take simple numbers: $S = 100$ and $D = 50$, so the linear regression slope is $100/50 = 2$. Figure 1 follows both slopes as λ grows.

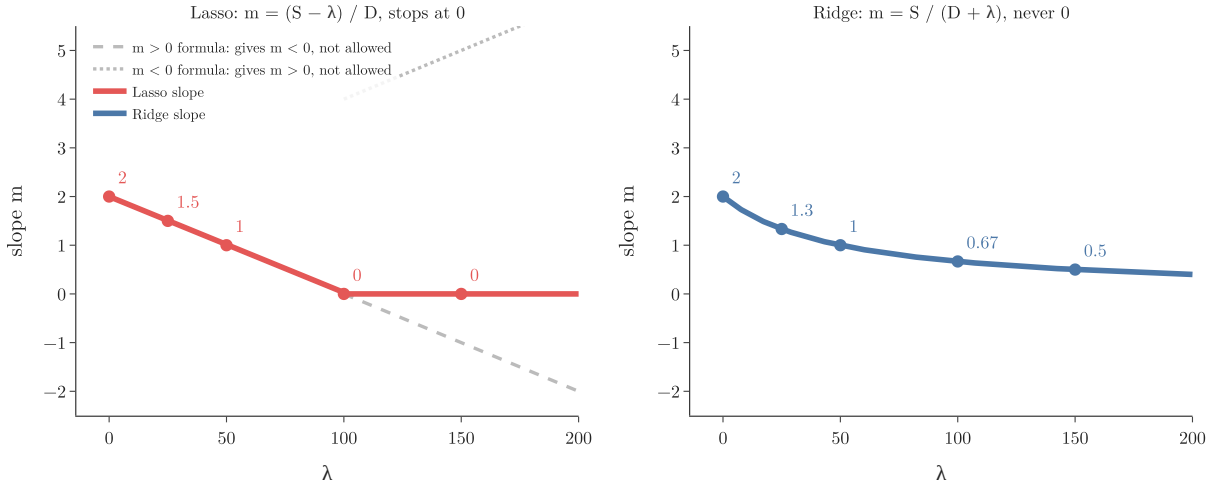


Figure 1: Slope against λ for Lasso and Ridge, with $S = 100$ and $D = 50$

λ	Lasso: $(100 - \lambda)/50$	Ridge: $100/(50 + \lambda)$
0	2	2
25	1.5	1.33
50	1	1
100	0	0.67

λ	Lasso: $(100 - \lambda)/50$	Ridge: $100/(50 + \lambda)$
150	0	0.5
1000	0	0.095

4.1 Why Lasso reaches 0

Key point: λ grows until it equals S ; then the numerator is 0.

At $\lambda = 100$, the numerator $S - \lambda$ is 0, so the slope is exactly 0.

4.2 Why Lasso stops at 0

Key point: Going past 0 would need the other case's formula, which pushes the slope back to the positive side.

At $\lambda = 150$, the positive-case formula gives $(100 - 150)/50 = -1$. The value -1 is negative, so the positive-case formula no longer applies (Figure 1, dashed). The negative-case formula would give $(100 + 150)/50 = 5$, which is positive, so it does not apply either (dotted).

Neither side has a valid answer, so the slope stays at $m = 0$. The same happens for every larger λ .

Extra: The same holds with a negative S . With $S = -100$ and $D = 50$, the slope starts at -2 , rises to -1 at $\lambda = 50$ and reaches 0 at $\lambda = 100$. At $\lambda = 150$ the negative-case formula would give $(-100 + 150)/50 = 1$, positive, so it does not apply, and the slope stays at 0.

4.3 Why Ridge never reaches 0

Key point: With λ only in the denominator, the fraction gets smaller but its top never changes.

An everyday picture: take a bill of 100 rupees. Subtracting a discount of 100 rupees brings it to exactly 0. Splitting it among more and more people only makes each share smaller; no share ever becomes 0. Lasso subtracts λ ; Ridge divides by something that grows with λ .

In Ridge's $S/(D + \lambda)$, the numerator stays at 100 however large λ gets. Even with $\lambda = 10,000,000$, the slope is $100/10,000,050$, tiny but not 0.

In short:

- **Lasso:** λ is in the **numerator**, so it can make the slope exactly 0.
- **Ridge:** λ is in the **denominator**, so it can only make the slope small.

5. The dead zone

Key point: For a fixed λ , every feature whose S lies between $-\lambda$ and λ gets a slope of exactly 0.

Figure 2 turns the view around: λ is fixed at 100, and the slope is drawn for every value of S .

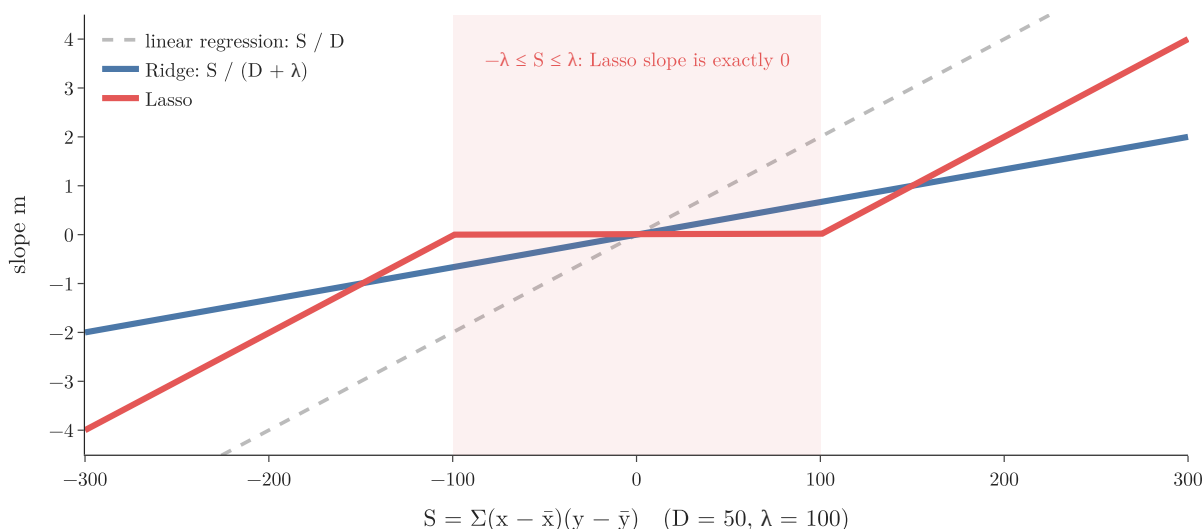


Figure 2: The slope against S for linear regression, Ridge and Lasso

- **Linear regression** (grey dashed): the slope is proportional to S .
- **Ridge** (blue): also proportional to S , just flatter. The Ridge slope is 0 only when S is exactly 0.
- **Lasso** (red): flat at 0 for every S between -100 and 100 , the shaded **dead zone**. Outside it, Lasso follows the linear regression line moved $\lambda/D = 2$ towards 0.

S measures how strongly the feature and the **target** (the output we predict) move together. So a feature with only a weak link to the target falls into the dead zone and is dropped. Dropping weak features in this way is the feature selection of the Lasso Note.

Think of λ as an entry fee. A feature's link with the target, $|S|$, must be larger than the fee to get any slope at all, and above the fee the feature keeps only what is left over.

Extra: Shrinking a value towards 0 by a fixed amount and setting it to 0 if it would cross 0 is called **soft thresholding**. With several features, Lasso has no single formula, but scikit-learn's method (coordinate descent) applies this same soft-threshold step to one coefficient at a time, over and over (ESL §3.8.6; scikit-learn docs, **Lasso**). The repeated soft-threshold step is why many coefficients land exactly on 0.

5.1 Ten features at once

Key point: On the 10-feature diabetes data, Lasso coefficients drop to exactly 0 one after another; Ridge coefficients only shrink.

Figure 3 grows λ for Ridge (left) and Lasso (right) on the diabetes data of the [Lasso Note](#) (same split). Watch the dots: a Lasso dot that reaches 0 turns into an open circle and stays there, while every Ridge dot keeps sliding towards 0 without arriving.

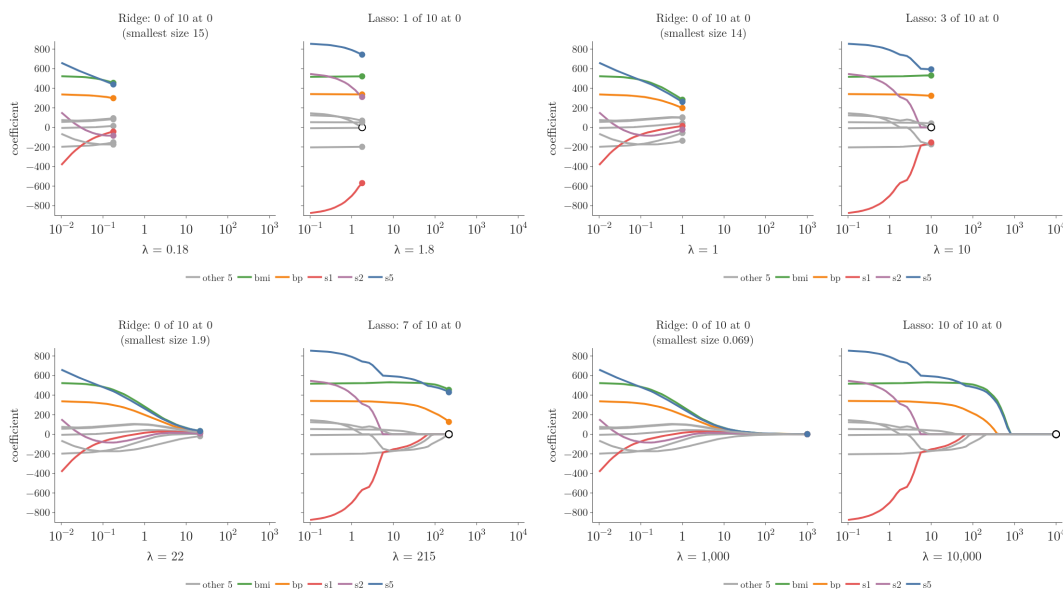


Figure 3: Ridge (left) and Lasso (right) coefficients of the 10 diabetes features as λ grows. Each panel has its own λ range, because Ridge shrinks faster on these features. An open circle marks a coefficient that is exactly 0

- **Lasso:** the coefficients reach 0 one at a time, bmi and s5 last; from $\lambda \approx 825$ all 10 are exactly 0.
- **Ridge:** at $\lambda = 1000$, all coefficients are tiny, but the smallest is still 0.069, not 0.

5.2 The picture: a diamond and a circle

Key point: Lasso's allowed region is a diamond with corners on the axes; the loss rings around the least-squares answer usually touch it first at a corner, where one coefficient is exactly 0.

The same result has a geometric picture (ESL §3.4.3, Figure 3.11). Ridge and Lasso can each be written as “make the loss as small as possible while the coefficients stay inside a budget”. With two coefficients and a budget t :

- **Lasso:** $|b_1| + |b_2| \leq t$, a diamond with its corners on the axes.
- **Ridge:** $b_1^2 + b_2^2 \leq t^2$, a circle.

The loss is a bowl over the (b_1, b_2) plane. Its lowest point is the linear regression (least-squares) answer, and points of equal loss form ellipses around it. Growing the ellipse until it first touches the allowed region gives the answer: the point of the region with the smallest loss.

Figure 4 does this on two features of the same diabetes training split, bmi and bp, each scaled to standard deviation 1, with the same budget $t = 15$ for both. Watch where each ring first meets its region.

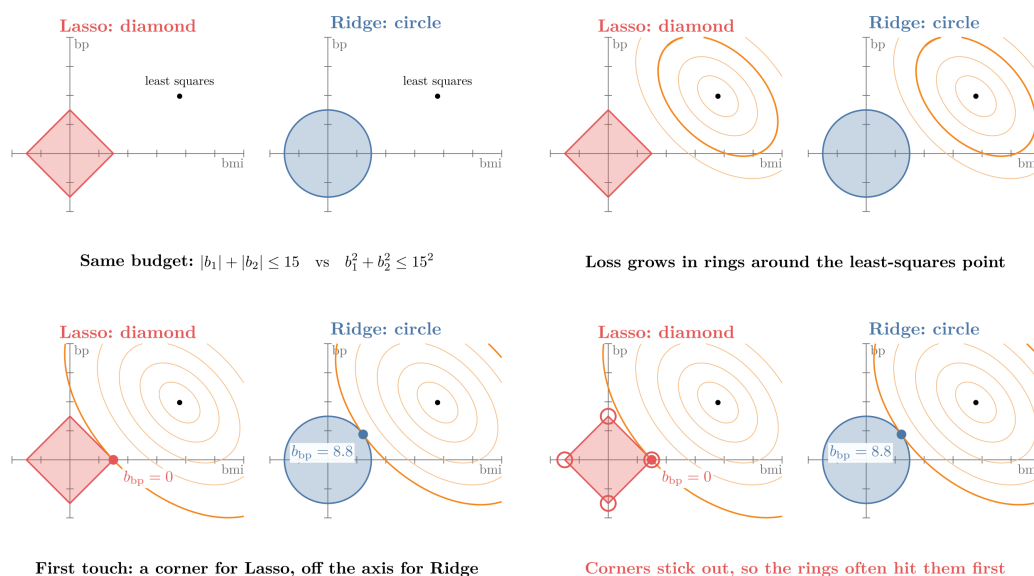


Figure 4: Lasso (left) and Ridge (right) with the same budget on two diabetes features. The loss ellipse grows from the least-squares point until it first touches each region: the diamond at its corner (bp coefficient exactly 0), the circle at a point where both coefficients are non-zero

- **Lasso:** the ring first touches the diamond exactly at its corner $(15, 0)$. The bp coefficient is exactly 0.
- **Ridge:** the circle has no corners, so the ring touches it at $(12.2, 8.8)$. Both coefficients are kept, just smaller than the least-squares $(37.9, 19.7)$.

The corners stick out towards the rings, so the rings often meet a corner first. With more features the diamond has many corners, edges and faces on which some coefficients are 0, so there are even more chances for exact zeros (ESL §3.4.3).

6. Checking with scikit-learn

Key point: On the 100-observation example, the formula and scikit-learn's Lasso give the same slopes, including the exact 0.

For the example of the earlier Notes, with 100 **observations** (records, one row of the data table each), $S = 2416.73$ and $D = 86.85$. So the slope reaches 0 at $\lambda = 2416.73$.

λ	Formula	scikit-learn Lasso
500	22.071	22.071
1000	16.313	16.313
2000	4.799	4.799
2416.73	0	0
3000	0	0

scikit-learn's Lasso divides the squared error by $2n$, so its `alpha` equals λ/n . With $n = 100$, the slope reaches 0 at `alpha = 24.17`: the value found in the Lasso Note.

Python: The one-feature Lasso slope.

```
def lasso_slope(S, D, lam):
    if S > lam:
        return (S - lam) / D
    if S < -lam:
        return (S + lam) / D
    return 0.0

lasso_slope(100, 50, 25)      # 1.5
lasso_slope(100, 50, 150)    # 0.0
```

7. Summary

	Ridge	Lasso
Slope, one feature	$S/(D + \lambda)$	$(S - \lambda)/D$, 0, or $(S + \lambda)/D$
Where λ appears	denominator	numerator
Large λ	slope small, never 0	slope exactly 0
Weak features	kept with small coefficients	dropped (dead zone)

- Sparsity means many coefficients exactly 0.
- The absolute value forces the Lasso formula into cases, and λ ends up subtracted from the numerator.
- Once $\lambda \geq |S|$, the slope is 0 and stays there.
- In pictures: the loss ellipse usually first touches Lasso's diamond at a corner, where a coefficient is exactly 0; Ridge's circle has no corners.

8. Sources

Built from

- CampusX, “Why Lasso Regression creates sparsity?”, YouTube, <https://www.youtube.com/watch?v=F>

Other references

- **ESL:** Hastie, T., Tibshirani, R. and Friedman, J. *The Elements of Statistical Learning*, 2nd ed. Springer, 2009. Section 3.4.3 and Figure 3.11 (the diamond and the disk), and Section 3.8.6, p. 93.
- **scikit-learn docs:** `sklearn.linear_model.Lasso` (coordinate descent), scikit-learn 1.9.

9. Key terms

Term	Meaning
Sparsity	Having many coefficients exactly equal to 0
Soft thresholding	Moving a value towards 0 by a fixed amount, and setting it to 0 if it would cross 0
Dead zone	The range of S for which the Lasso slope is exactly 0
Coordinate descent	An optimisation method that updates one coefficient at a time; used by scikit-learn's Lasso